

Natnael Messay

Software Developer

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Performance-driven Full Stack Developer focused on building scalable and efficient applications. Proficient in Data Structures and Algorithms for optimized, high-performing code. Detail-oriented and committed to clean, maintainable, and well-documented practices. Experienced in creating responsive user interfaces and robust backends. Collaborative, adaptable, and passionate about delivering reliable, user-focused software.

Skills

Front-End

Html, CSS, Tailwind CSS, Javascript, React.js, Next.js

Database

MYSQL, MongoDB, PostgreSQL

Version Control

Git, Github

Back-End

Node.js, express.js, Nest.js, Go, FastApi

Deployment

Vercel, Render, Heroku, Netlify

Professional Experience

FullStack Developer, Infnova technologies

09/2025 – 02/2026

Adiss Ababa, Ethiopia

- **Full-Stack Architecture:** Developed high-performance systems using **NestJS** and **Next.js**, utilizing **Prisma** with **PostgreSQL** to automate complex cloud workflows and improve data reliability.
- **Security Engineering:** Eliminated 100% of credential leaks by routing **Google Cloud** requests through a secure **Serverless Proxy** layer protected by **NestJS** authentication guards.
- **System Throughput:** Built background processing pipelines using **Redis**, increasing simultaneous user tasks by 60% through optimized **PostgreSQL** connection pooling and query management.
- **Performance Optimization:** Achieved 95+ Lighthouse scores by implementing **ISR** and **PostgreSQL** indexing, resulting in instant page loads and sub-second data updates via **Prisma**.

Backend Developer, 4K Labs

09/2023 – 04/2024

Adiss Ababa, Ethiopia

- **Full-Stack Orchestration:** Converted a static website into a dynamic **MERN** platform with modular **Node.js** APIs, optimized **MongoDB** queries, and custom **indexing** for efficient schema design.
- **Security & Performance:** Implemented **in-memory caching**, **JWT** authentication, and **RBAC** middleware to ensure secure departmental data isolation and robust error handling.
- **Modernized Workflows:** Migrated legacy data to **MongoDB** and refactored backend logic to support **Agile** workflows, automated testing.

Projects

News Recommendation Website 🔗

- **Personalized News Recommender:** Developed a full-stack application using **FastAPI** and **React** that fetches and stores news articles in **PostgreSQL**, delivering personalized content based on user interests.
- **NLP-Driven Logic:** Leveraged **spaCy** for Named Entity Recognition (NER) and keyword extraction to transform unstructured article descriptions into searchable metadata.
- **Modern Architecture:** Engineered a modular system featuring a RESTful API and a **Tailwind CSS** frontend, utilizing vector-based keyword matching to rank and recommend the top 5 most relevant articles.

Hospital Management Sytem 🔗

- **Full-Stack Clinical Orchestration:** Architected a **MERN** (MongoDB, Express, React, Node.js) hospital platform featuring **Role-Based Access Control (RBAC)** and **JWT-secured** data tunnels to synchronize workflows between Patients, Specialists, and Pharmacists.
- **Real-Time Persistent Event Engine:** Eliminated redundant API polling by integrating **Socket.io** for instant clinical alerts, backed by a **MongoDB** notification store to ensure 100% message delivery for offline medical staff.
- **High-Performance Frontend Engineering:** Developed a modular **React** interface utilizing **React Query** for server-state caching and **Virtual Scrolling**, reducing DOM overhead and API latency to deliver a seamless 60FPS experience for large-scale medical records.

Education

Bachelor of Science in Computer Engineering, Addis Ababa Universty

01/2021 – 06/2025

- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Database Systems, Object-Oriented Programming (OOP), Computer Security, and System Architecture.

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